

Disease Knowledge Base

Structured health data for a RAG / retrieval project

This document is designed as a compact knowledge base for Retrieval-Augmented Generation (RAG). Facts are summarized from World Health Organization (WHO) fact sheets and are intended for educational/information-retrieval use, not diagnosis or treatment decisions. Disease statistics refer to the year stated in each entry and should not be interpreted as current-year counts.

Dataset size: 16 disease/condition entries

Recommended retrieval fields: name, type, cause, spread, symptoms, risk factors, prevention, treatment, data, source.

1. Tuberculosis (TB)

Type	Infectious • Bacterial
Cause	Mycobacterium tuberculosis bacteria, most often affecting the lungs.
Transmission / spread	Airborne transmission when a person with infectious TB coughs, sneezes or spits.
Symptoms	Prolonged cough, chest pain, weakness, fatigue, weight loss, fever and night sweats; cough may contain blood.
Risk factors	Higher risk with diabetes, weakened immunity, undernutrition, tobacco use and harmful alcohol use.
Prevention	Early testing and treatment, respiratory precautions, contact investigation, and BCG vaccination where recommended.
Treatment / management	Curable with appropriate multidrug antibiotic treatment under medical supervision.
Key data	WHO estimated 10.7 million people fell ill with TB and 1.23 million died in 2024.

Source: WHO, Tuberculosis fact sheet (24 Mar 2026). <https://www.who.int/en/news-room/fact-sheets/detail/tuberculosis>

2. Malaria

Type	Infectious • Parasitic
Cause	Plasmodium parasites, with P. falciparum and P. vivax among the most important human species.
Transmission / spread	Usually through bites of infected female Anopheles mosquitoes; rarely through contaminated blood or needles.
Symptoms	Fever, chills and headache; severe disease can cause impaired consciousness, seizures, breathing difficulty, jaundice or abnormal bleeding.
Risk factors	Young children, pregnant people, travellers and people with HIV/AIDS can be at increased risk of severe disease.
Prevention	Mosquito nets, repellents, protective clothing, vector control and preventive medicines when indicated for travel or risk groups.
Treatment / management	Treat promptly with an appropriate antimalarial regimen; severe malaria is a medical emergency.
Key data	WHO estimated 282 million malaria cases and 610,000 deaths globally in 2024.

Source: WHO, Malaria fact sheet (4 Dec 2025). <https://www.who.int/news-room/fact-sheets/detail/malaria>

3. Dengue

Type	Infectious • Viral
Cause	Dengue virus (DENV).
Transmission / spread	Bites of infected Aedes mosquitoes, especially Aedes aegypti.

Symptoms	High fever, severe headache, pain behind the eyes, muscle/joint pain, nausea, vomiting and rash. Severe dengue may cause abdominal pain, persistent vomiting, bleeding, weakness or shock.
Risk factors	People infected previously with a different dengue serotype have increased risk of severe dengue.
Prevention	Avoid mosquito bites, remove standing water and use community vector-control measures.
Treatment / management	Supportive medical care and monitoring; there is no specific antiviral treatment for dengue.
Key data	WHO estimates 100–400 million dengue infections occur each year; 14.6 million reported cases occurred in 2024.

Source: WHO, Dengue and severe dengue fact sheet (21 Aug 2025). <https://www.who.int/news-room/fact-sheets/detail/dengue-and-severe-dengue>

4. Rabies

Type	Infectious • Viral • Zoonotic
Cause	Rabies virus, which affects the central nervous system.
Transmission / spread	Usually through saliva from infected mammals, especially dog bites or scratches.
Symptoms	Early illness may include fever and abnormal sensations near the wound; later disease can cause agitation, difficulty swallowing, hydrophobia, paralysis and coma.
Risk factors	People exposed to potentially rabid animals without prompt post-exposure care.
Prevention	Vaccination of dogs, avoiding animal bites, immediate wound washing and prompt post-exposure prophylaxis after risky exposure.
Treatment / management	Once clinical symptoms appear, rabies is virtually 100% fatal; prompt post-exposure prophylaxis can prevent disease before symptoms develop.
Key data	WHO reports that dogs cause up to 99% of human rabies cases and that rabies causes tens of thousands of deaths annually.

Source: WHO, Rabies fact sheet (17 Sep 2026). <https://www.who.int/news-room/fact-sheets/detail/rabies>

5. Measles

Type	Infectious • Viral
Cause	Measles virus.
Transmission / spread	Highly contagious airborne/respiratory transmission from infected people.
Symptoms	High fever, cough, runny nose, red/watery eyes, small white spots inside the cheeks and a widespread rash.
Risk factors	Unvaccinated or under-vaccinated people, especially young children and people with malnutrition or weakened immunity.
Prevention	Routine measles vaccination is the main prevention strategy.

Treatment / management	No specific antiviral treatment; supportive care and management of complications are used.
Key data	WHO estimated about 95,000 measles deaths globally in 2024.

Source: WHO, Measles fact sheet (15 Jul 2026). <https://www.who.int/news-room/fact-sheets/detail/measles>

6. HIV infection / AIDS

Type	Infectious • Viral
Cause	Human immunodeficiency virus (HIV), which progressively damages the immune system without effective treatment.
Transmission / spread	Exchange of infected blood, semen, vaginal fluids or breast milk; transmission can also occur from mother to child during pregnancy/delivery.
Symptoms	Early infection may be asymptomatic or flu-like; later untreated infection can cause weight loss, fever, diarrhoea, cough and opportunistic infections.
Risk factors	Exposure to infected body fluids and situations with increased HIV transmission risk.
Prevention	Condoms and other barrier methods, testing, PrEP where appropriate, safe injection practices, treatment that suppresses viral load, and prevention of mother-to-child transmission.
Treatment / management	Antiretroviral therapy (ART) controls HIV and can allow long, healthy lives, but does not currently cure HIV.
Key data	WHO estimated 41.0 million people were living with HIV at the end of 2025.

Source: WHO, HIV and AIDS fact sheet (27 Jul 2026). <https://www.who.int/en/news-room/fact-sheets/detail/hiv-aids>

7. Hepatitis B

Type	Infectious • Viral
Cause	Hepatitis B virus (HBV), which infects the liver.
Transmission / spread	Mother-to-child transmission during birth, early childhood transmission, and contact with infected blood or body fluids including through sex or unsafe injections.
Symptoms	Often asymptomatic; possible fatigue, nausea, abdominal discomfort, dark urine and jaundice. Chronic infection can cause cirrhosis and liver cancer.
Risk factors	Perinatal exposure, unsafe injections, unprotected sexual exposure and other blood/body-fluid exposures.
Prevention	Highly effective hepatitis B vaccination, safe injections and blood screening, and prevention of mother-to-child transmission.
Treatment / management	Chronic hepatitis B can be managed with antiviral medicines in appropriate patients; medical monitoring is important.
Key data	WHO estimated 240 million people had chronic hepatitis B infection in 2024.

Source: WHO, Hepatitis B fact sheet (28 Jul 2026). <https://www.who.int/news-room/fact-sheets/detail/hepatitis-b>

8. Hepatitis C

Type	Infectious • Viral
Cause	Hepatitis C virus (HCV), causing acute or chronic liver inflammation.

Transmission / spread	Primarily contact with infected blood, including unsafe injections/procedures, unscreened blood and shared needles.
Symptoms	Often no symptoms; possible fever, fatigue, poor appetite, nausea, abdominal pain, dark urine and jaundice.
Risk factors	Blood exposure, unsafe medical injections/procedures and sharing needles/syringes.
Prevention	Safe injections and blood screening; avoid sharing equipment that may be contaminated with blood. No effective vaccine currently exists.
Treatment / management	Direct-acting antiviral medicines can cure more than 95% of people with hepatitis C infection.
Key data	WHO estimates about 47 million people have chronic hepatitis C infection globally.

Source: WHO, Hepatitis C fact sheet (28 Jul 2026). <https://www.who.int/news-room/fact-sheets/detail/hepatitis-c>

9. Cholera

Type	Infectious • Bacterial
Cause	Toxigenic <i>Vibrio cholerae</i> , mainly serogroups O1 and O139 in outbreaks.
Transmission / spread	Usually through food or water contaminated with feces containing the bacteria.
Symptoms	Acute watery diarrhoea, vomiting and dehydration; severe dehydration can become fatal within hours if untreated.
Risk factors	Unsafe water, inadequate sanitation and hygiene, and living in or travelling to areas with outbreaks or poor WASH infrastructure.
Prevention	Safe drinking water, sanitation, hand hygiene, food safety and vaccination in appropriate public-health settings.
Treatment / management	Rapid oral rehydration is central; severe cases may require intravenous fluids and antibiotics under medical care.
Key data	Most infections are mild or asymptomatic, but severe cholera can progress rapidly.

Source: WHO, Cholera fact sheet. <https://www.who.int/en/news-room/fact-sheets/detail/cholera>

10. Type 2 Diabetes

Type	Noncommunicable • Metabolic
Cause	A chronic disorder in which the body does not use insulin effectively and/or cannot produce enough insulin to maintain normal blood glucose.
Transmission / spread	Not contagious.
Symptoms	May be absent for years; possible increased thirst, frequent urination, fatigue, blurred vision and unexplained weight change.
Risk factors	Overweight/obesity, physical inactivity, unhealthy diet, age, family history and other metabolic risk factors.

Prevention	Healthy diet, regular physical activity, healthy body weight and avoiding tobacco can reduce or delay risk.
Treatment / management	Lifestyle measures, blood-glucose monitoring and medicines as prescribed; management of cardiovascular and other complications is important.
Key data	WHO reports that the number of people living with diabetes rose to about 830 million in 2022.

Source: WHO, Diabetes fact sheet (14 Nov 2024). <https://www.who.int/en/news-room/fact-sheets/detail/diabetes>

11. Hypertension

Type	Noncommunicable • Cardiovascular
Cause	Persistently elevated blood pressure; causes are often multifactorial.
Transmission / spread	Not contagious.
Symptoms	Usually no symptoms. Very high blood pressure can be associated with severe headache, chest pain, breathing difficulty, visual changes or neurological symptoms.
Risk factors	Older age, family history, high-salt diet, excess weight, physical inactivity, tobacco, harmful alcohol use and some diseases/medicines.
Prevention	Reduce salt, eat a balanced diet, exercise regularly, maintain a healthy weight, avoid tobacco and limit alcohol; measure blood pressure.
Treatment / management	Lifestyle measures and antihypertensive medicines when indicated, with regular monitoring.
Key data	WHO estimates the number of adults with hypertension reached 1.4 billion in 2024.

Source: WHO, Hypertension fact sheet. <https://www.who.int/news-room/fact-sheets/detail/hypertension>

12. Stroke

Type	Noncommunicable • Neurological emergency
Cause	Interruption of brain blood flow from a blockage (ischaemic stroke) or bleeding (haemorrhagic stroke).
Transmission / spread	Not contagious.
Symptoms	Sudden facial droop, arm weakness, speech difficulty, vision loss, dizziness or loss of coordination.
Risk factors	High blood pressure, smoking, high cholesterol, diabetes, obesity, physical inactivity, unhealthy diet, harmful alcohol use, atrial fibrillation and age.
Prevention	Control blood pressure, avoid tobacco, exercise, maintain a healthy diet, manage diabetes/cholesterol and address atrial fibrillation when present.
Treatment / management	Emergency evaluation is essential; treatment depends on stroke type and timing and may include clot-busting therapy, thrombectomy, surgery and rehabilitation.
Key data	WHO estimated 93.8 million people living with stroke and 11.9 million new strokes in 2021.

Source: WHO, Stroke fact sheet (19 Dec 2025). <https://www.who.int/news-room/fact-sheets/detail/stroke>

13. Asthma

Type	Noncommunicable • Chronic respiratory
Cause	Chronic airway inflammation and variable narrowing; triggers vary between individuals.
Transmission / spread	Not contagious.

Symptoms	Cough, wheezing, shortness of breath and chest tightness; symptoms can fluctuate and flare.
Risk factors	Family/genetic susceptibility, allergens, air pollution, tobacco smoke and occupational or respiratory exposures.
Prevention	Identify and reduce individual triggers, avoid tobacco smoke, reduce relevant exposures and use prescribed controller medication.
Treatment / management	Inhaled medicines can control symptoms and reduce exacerbations; treatment should be individualized by a clinician.
Key data	WHO estimated 363 million people had asthma in 2023 and 442,000 deaths occurred that year.

Source: WHO, Asthma fact sheet (28 Apr 2026). <https://www.who.int/en/news-room/fact-sheets/detail/asthma>

14. COPD

Type	Noncommunicable • Chronic respiratory
Cause	Persistent airflow limitation caused by lung damage, commonly associated with tobacco smoke and air pollution.
Transmission / spread	Not contagious.
Symptoms	Chronic cough, phlegm, breathlessness, wheezing and tiredness; flare-ups can worsen symptoms.
Risk factors	Smoking, second-hand smoke, household air pollution, occupational exposures and other lung irritants.
Prevention	Do not smoke, avoid second-hand smoke and reduce harmful air-pollution exposures; vaccination can reduce some respiratory infections.
Treatment / management	Smoking cessation, inhaled medicines, oxygen for selected patients and pulmonary rehabilitation can improve symptoms and function.
Key data	WHO reports 3.4 million COPD deaths in 2023, about 6% of global deaths.

Source: WHO, COPD fact sheet (10 Jun 2026).
<https://www.who.int/news-room/fact-sheets/detail/chronic-obstructive-pulmonary-disease-%28copd%29>

15. Cancer

Type	Noncommunicable • Malignant disease
Cause	Abnormal cells grow uncontrollably and can invade nearby tissue or spread to other organs; causes differ by cancer type.
Transmission / spread	Cancer itself is not generally contagious, although some infections that increase cancer risk can be transmissible.
Symptoms	Depend on cancer type and site; possible unexplained weight loss, persistent lump, bleeding, prolonged cough, bowel changes or persistent pain.
Risk factors	Tobacco, alcohol, unhealthy diet, physical inactivity, obesity, air pollution, ultraviolet radiation and some chronic infections.

Prevention	Avoid tobacco, maintain healthy weight, physical activity, healthy diet, reduce alcohol, protect against UV exposure and use recommended HPV/hepatitis B vaccination.
Treatment / management	May include surgery, radiotherapy, chemotherapy, targeted therapy, immunotherapy or combinations depending on cancer type and stage.
Key data	WHO reports nearly 10 million cancer deaths worldwide in 2024.

Source: WHO, Cancer fact sheet (3 Jul 2026). <https://www.who.int/en/news-room/fact-sheets/detail/cancer>

16. Heart Attack / Ischaemic Heart Disease

Type	Noncommunicable • Cardiovascular emergency
Cause	Usually atherosclerotic plaque in a coronary artery ruptures and a clot suddenly reduces or blocks blood flow to heart muscle.
Transmission / spread	Not contagious.
Symptoms	Chest pressure/pain, shortness of breath, sweating, nausea, light-headedness or pain/discomfort spreading to the arm, shoulder, jaw, neck or back. Symptoms can be atypical.
Risk factors	Smoking, high blood pressure, high LDL cholesterol, diabetes, obesity, physical inactivity, unhealthy diet and age.
Prevention	Do not smoke, control blood pressure/cholesterol/diabetes, exercise regularly, eat a balanced diet and follow prescribed preventive treatment.
Treatment / management	Emergency medical care; hospital treatment may restore blood flow using medicines and/or procedures such as coronary intervention.
Key data	WHO reports ischaemic heart disease caused about 9 million deaths in 2021 and remains the leading cause of death worldwide.

Source: WHO, Heart attack fact sheet (10 Sep 2026). <https://www.who.int/news-room/fact-sheets/detail/heart-attack>

RAG Implementation Notes

- Store each disease entry as one document/chunk, or split by field if your retriever performs field-level search.
- Use the disease name and aliases as metadata (for example, TB = tuberculosis; COPD = chronic obstructive pulmonary disease).
- Keep the year attached to every statistic because global disease estimates change over time.
- For production medical applications, add a clinical review workflow and prefer authoritative, current sources.
- Do not use the dataset to diagnose an individual or recommend a personalized treatment plan.

Primary reference organization: World Health Organization (WHO)

All source URLs are included with the individual disease records above. Accessed/compiled October 2026.